Project 9: Highly Available Static Website on AWS using EC2, ELB (ALB) & Auto Scaling

Project Description

This project demonstrates deployment of a highly available static website on AWS using Amazon EC2, Application Load Balancer (ALB), Auto Scaling Group (ASG), and a custom VPC. A master EC2 instance was configured with Apache HTTP Server and the website files. An Amazon Machine Image (AMI) was created and used in a Launch Template. Auto Scaling launches identical instances across multiple Availability Zones while the ALB distributes incoming traffic and performs health checks.

Prerequisites

- AWS Account
- Custom VPC with public subnets in two Availability Zones
- Internet Gateway and Route Table
- Security Groups
- EC2 instance with Apache
- AMI and Launch Template
- Application Load Balancer
- Target Group
- Auto Scaling Group

Implementation Steps

1. Created a custom VPC with two public subnets across different Availability Zones.
2. Configured Internet Gateway and Route Table for internet connectivity.
3. Created Security Groups allowing HTTP (80) and SSH (22).
4. Launched a master EC2 instance and installed Apache HTTP Server.
5. Uploaded the CloudWithKushagra website to /var/www/html and verified it using the EC2 public IP.
6. Created an Amazon Machine Image (AMI) from the configured EC2 instance.
7. Created a Launch Template using the custom AMI.
8. Created a Target Group with HTTP health checks.
9. Created an Internet-facing Application Load Balancer and associated it with the Target Group.
10. Created an Auto Scaling Group using the Launch Template and attached it to the Target Group.
11. Verified healthy targets and tested website accessibility using the ALB DNS name.

Result

The static website was successfully deployed in a highly available architecture. Traffic is distributed through the Application Load Balancer, Auto Scaling automatically maintains the desired number of instances, and the application remains available even if an instance fails.

Project Screenshots

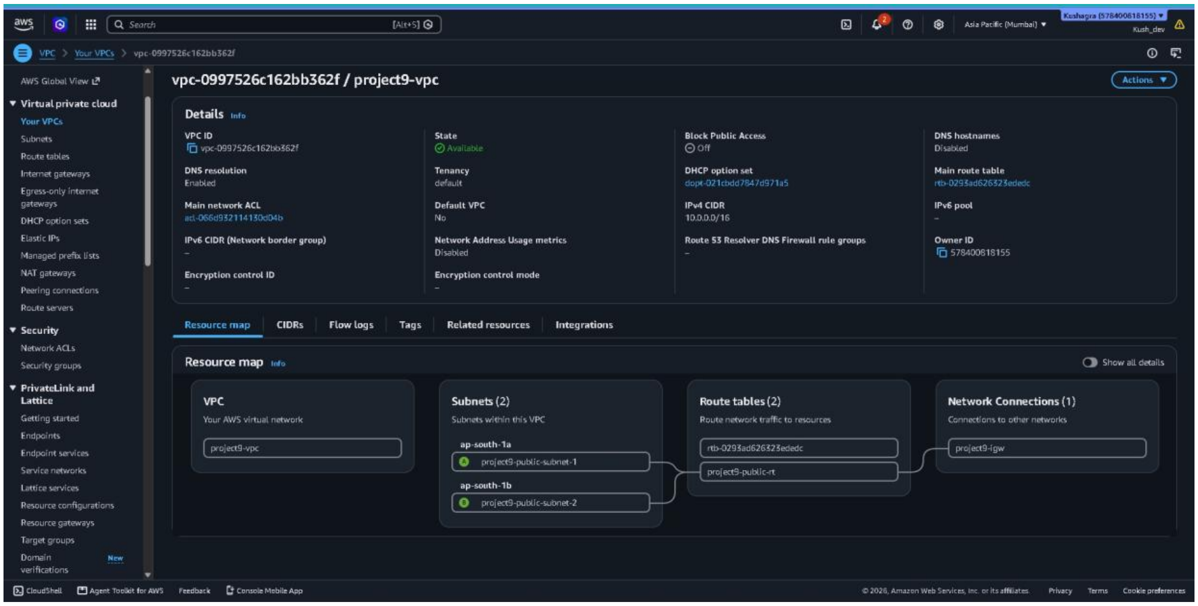

Figure 1Custom VPC (project9-vpc) Created Successfully

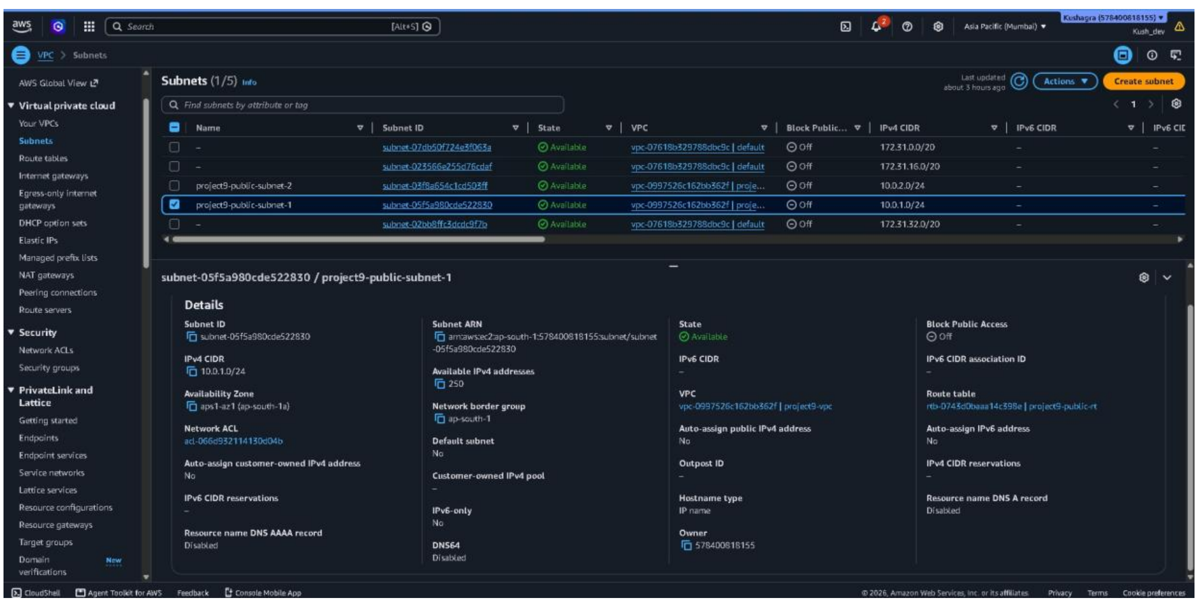

Figure 2: Public Subnets Created in Two Availability Zones

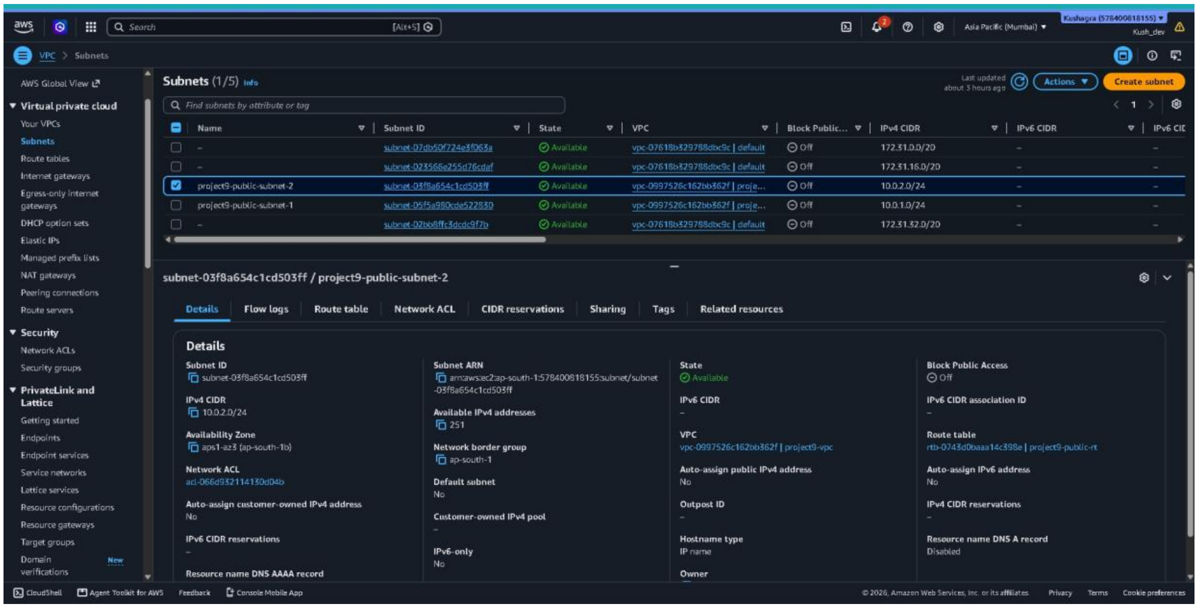

Figure 3: Public Subnets Created in Two Availability Zones

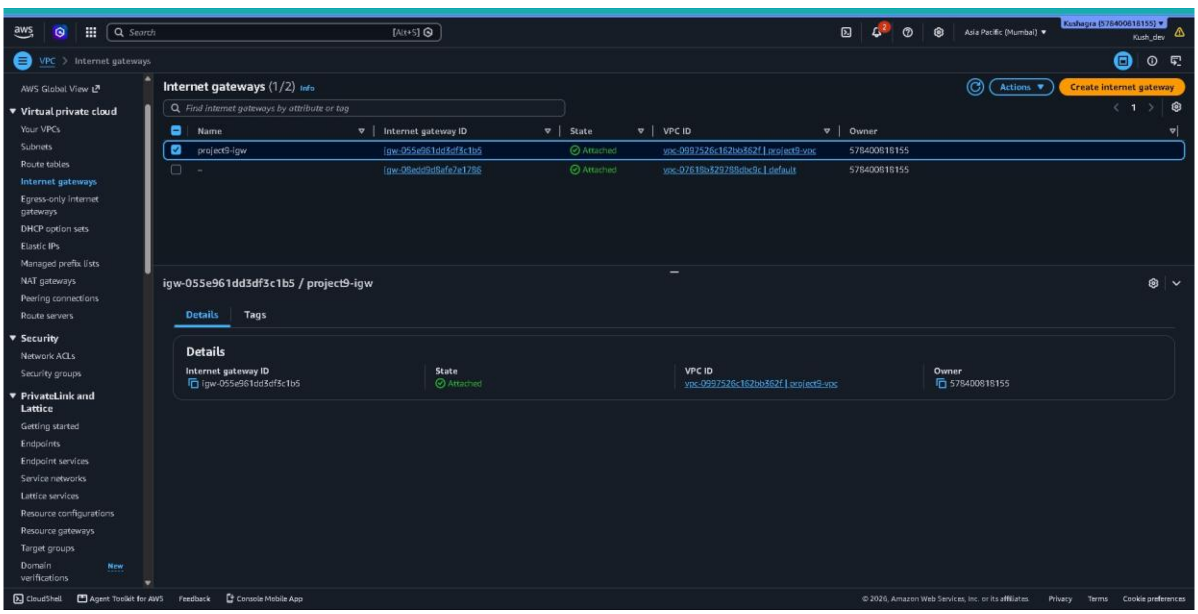

Figure 4: Internet Gateway Attached to the VPC

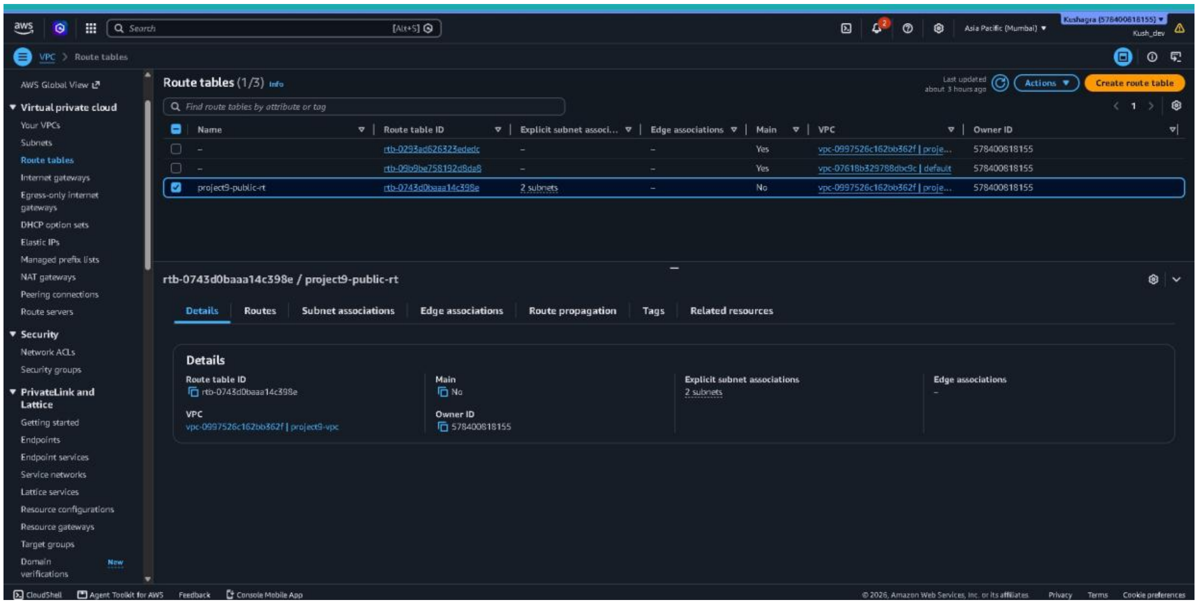

Figure 5: Public Route Table Configured with Internet Gateway

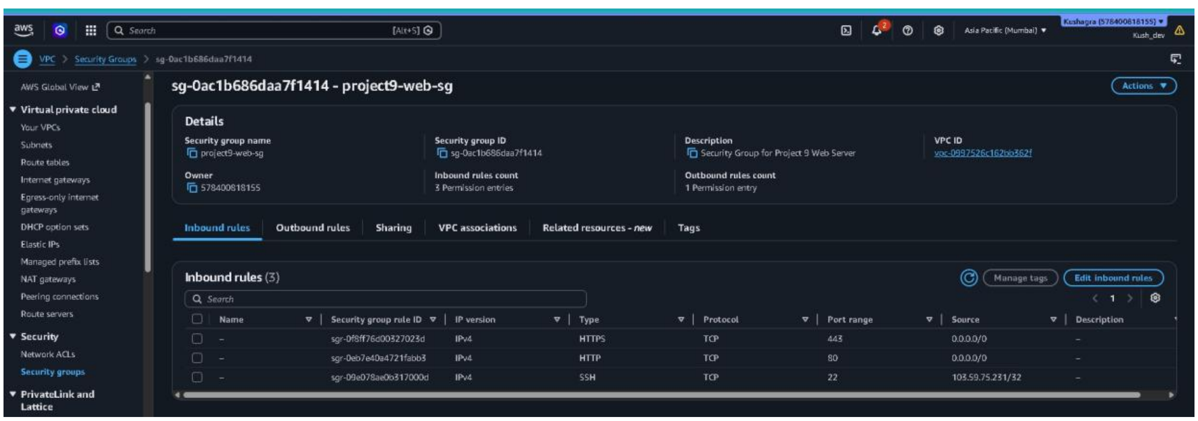

Figure 6: Security Group (project9-web-sg) Configuration

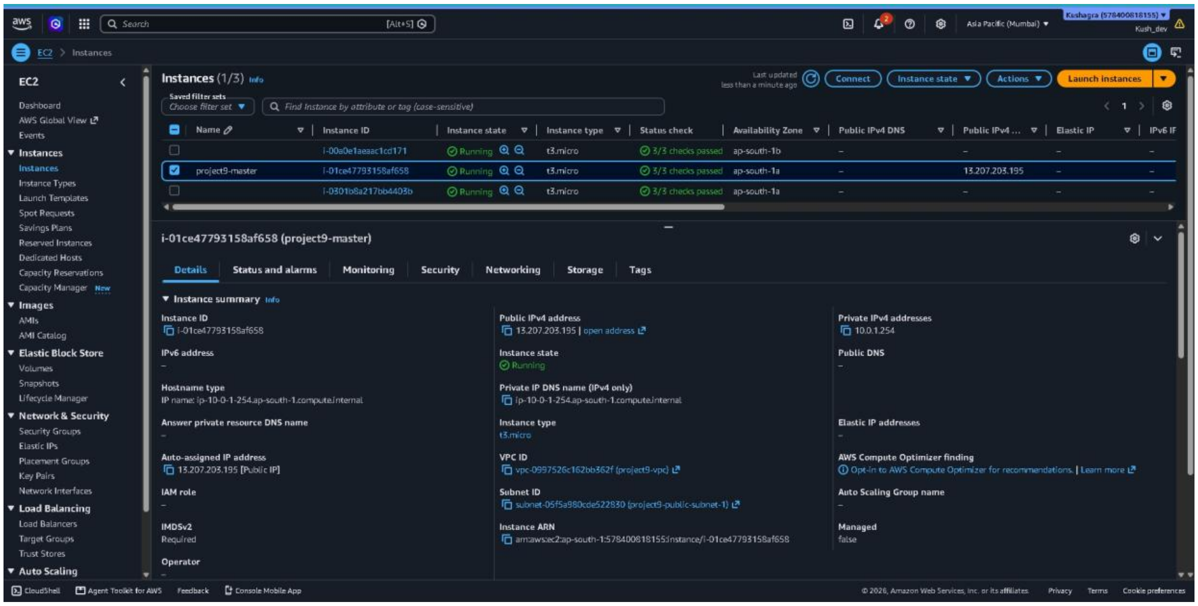

Figure 7: EC2 Master Instance Running

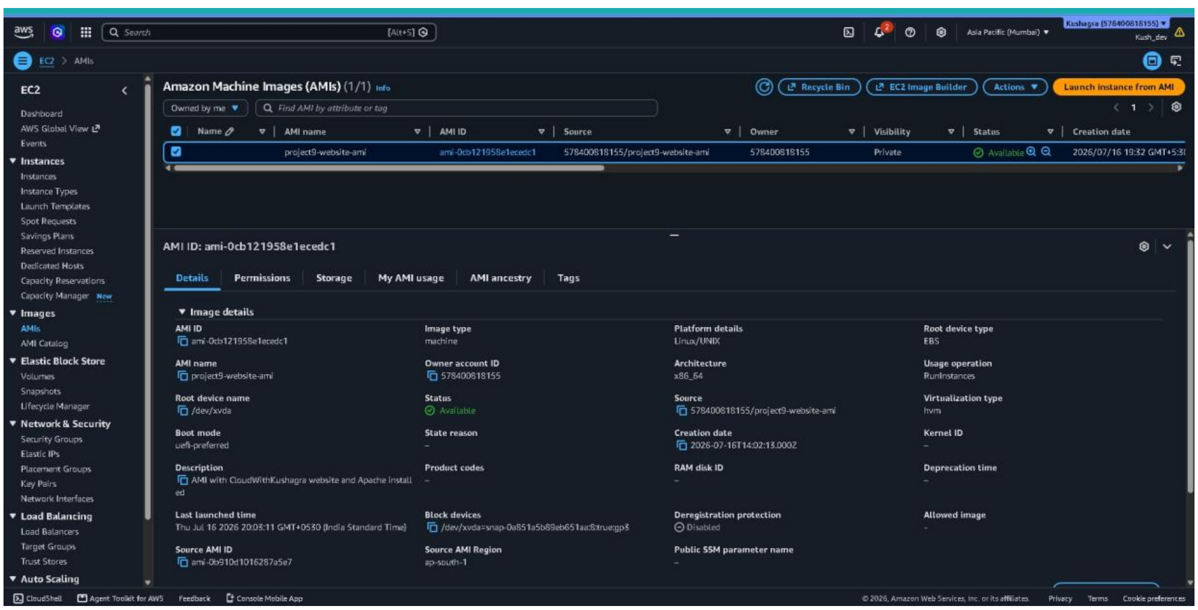

Figure 8: Amazon Machine Image (AMI) Created Successfully

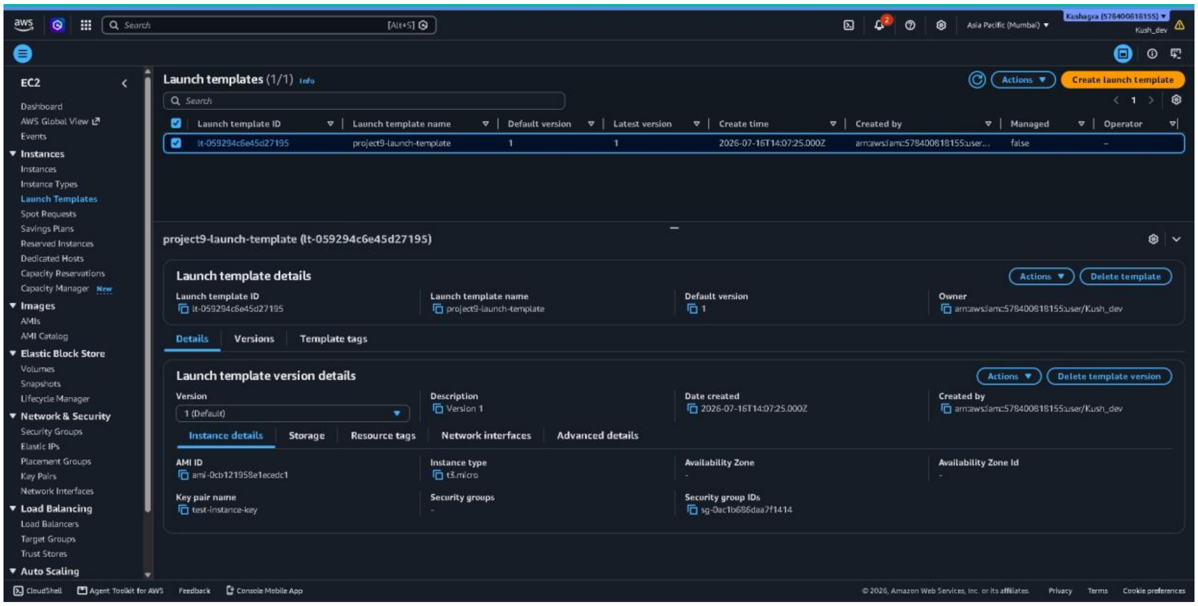

Figure 9: Launch Template Using Custom AMI

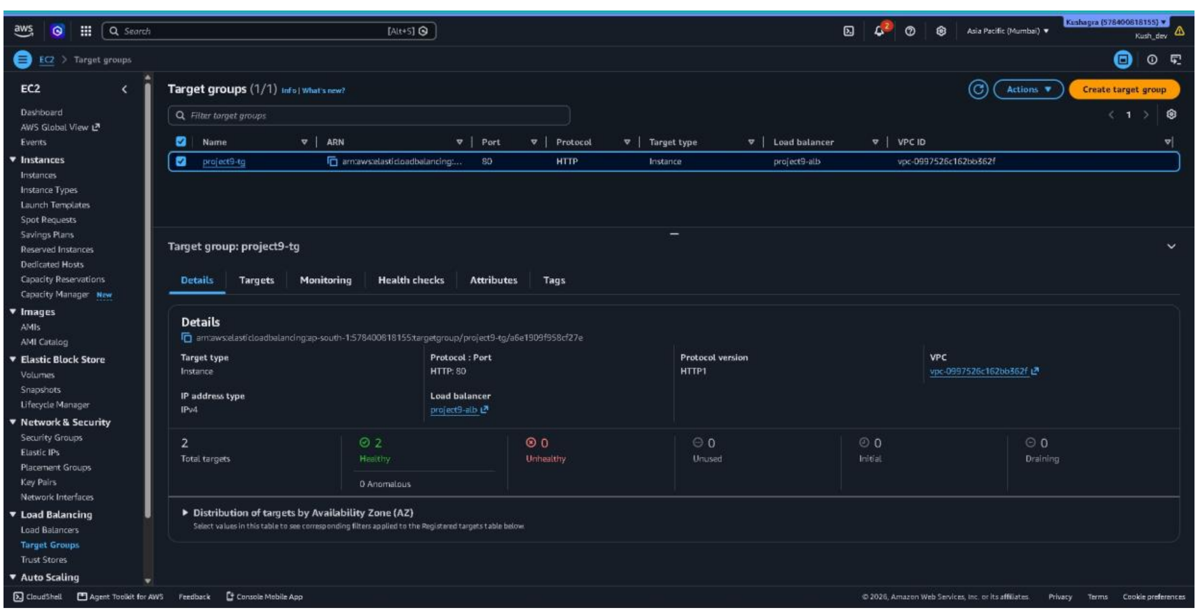

Figure 10: Target Group with Healthy Registered Targets

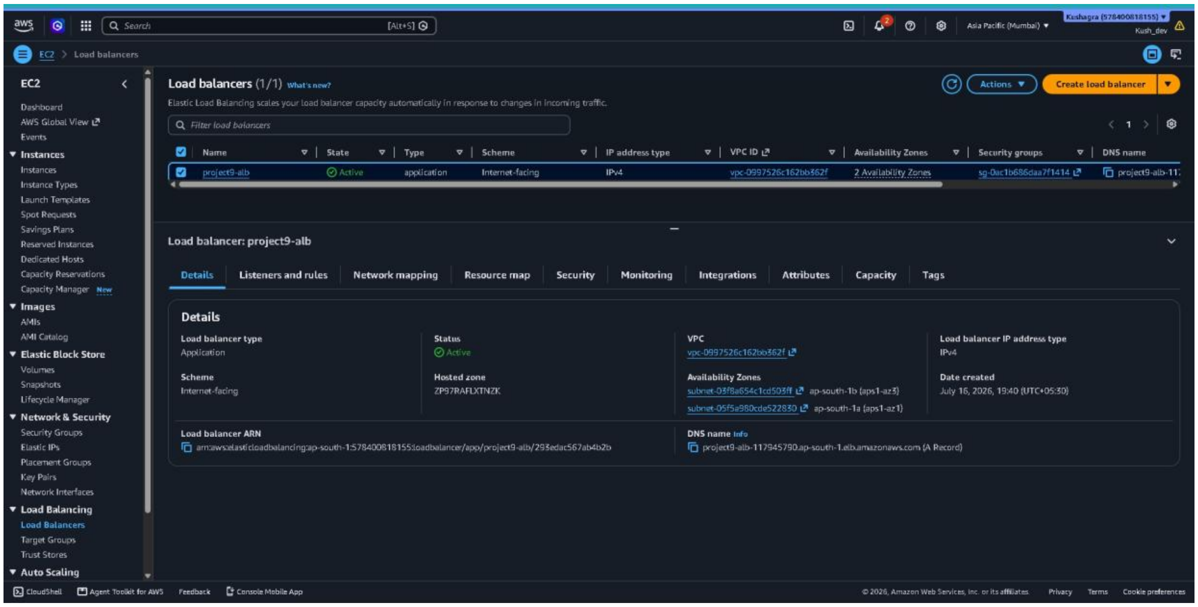

Figure 11: Application Load Balancer Configuration

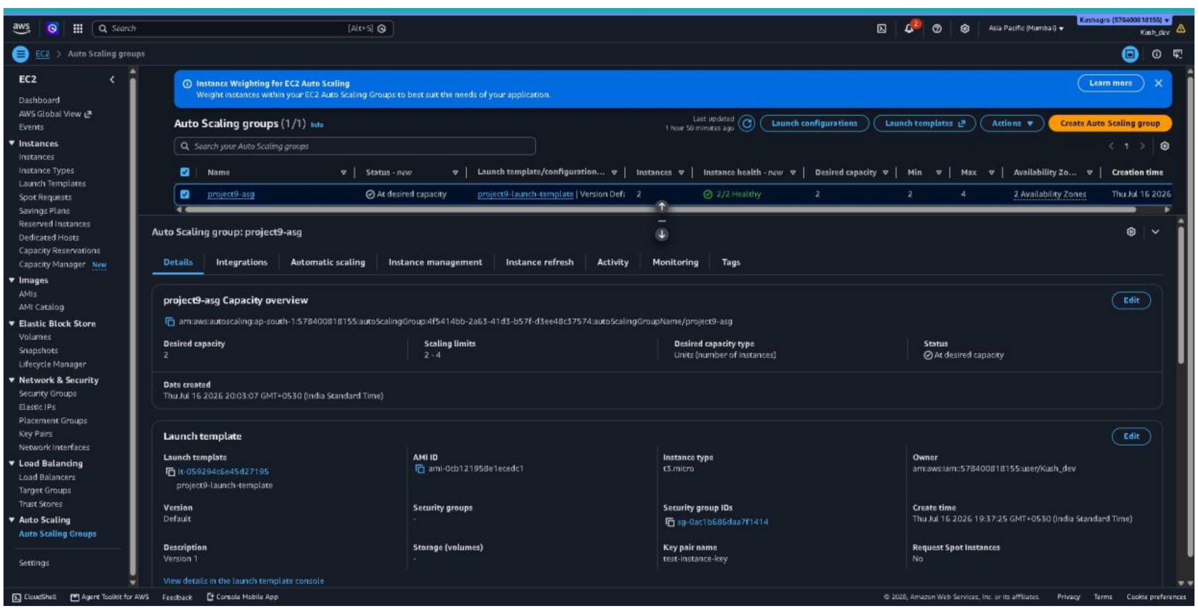

Figure 12: Auto Scaling Group Configuration

Figure 13: Auto Scaling Instances Running Successfully

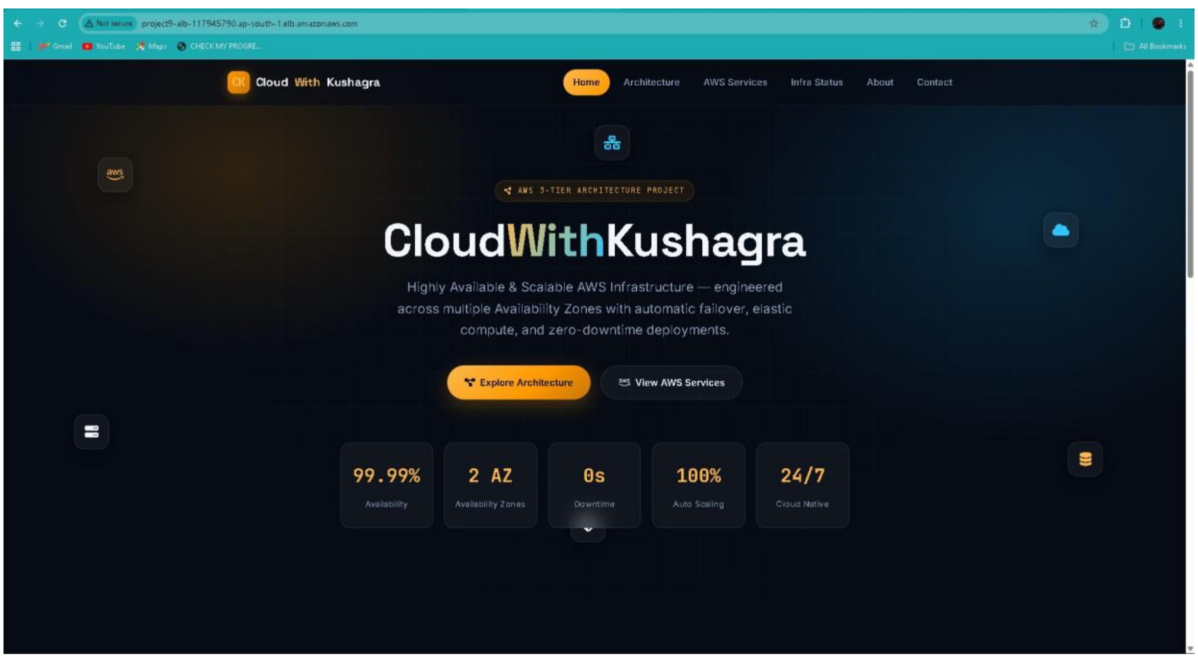

Figure 14: Website Successfully Accessed Through Application Load Balancer

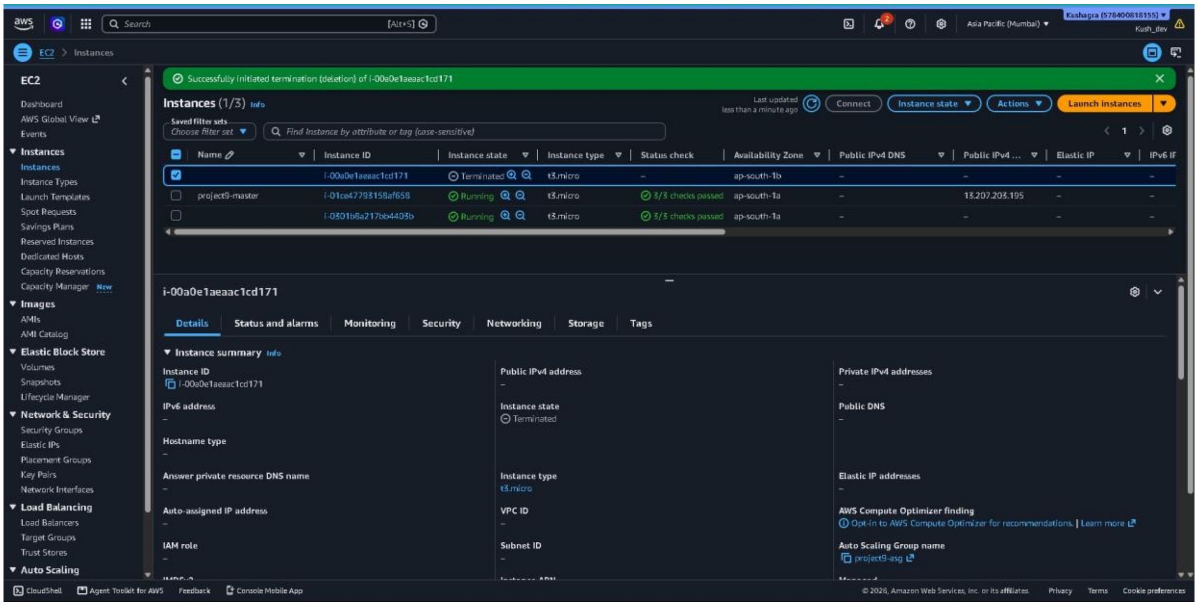

Figure 15: High Availability Achieved Using ALB and Auto Scaling

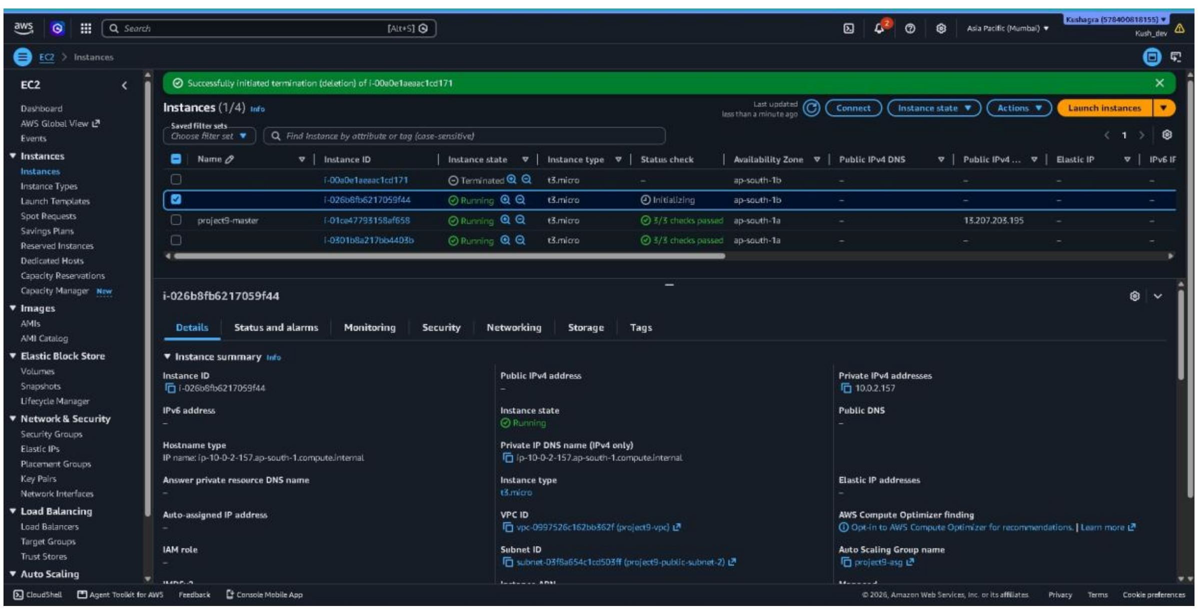

Figure 16: High Availability Achieved Using ALB and Auto Scaling